

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)
Department of Computer Science and Engineering
CO4_AT1 — RELATIONSHIP VISUALIZATION LAB

Course Code:	DSA06	Course Title:	Data Handling and Visualization
CO Assessed:	CO4 – Analysis (BL4)	Assessment:	Assessment Tool 1 – Relationship Visualization Lab
Weightage:	20% of CIA	Total Marks:	25 (5 Questions × 5 Marks)
CO4 Statement:	<i>Analyze relationships between variables using scatterplots, correlograms, PCA, bubble charts, and network graphs to construct and interpret appropriate visualizations.(BL4)</i>		
Student Name:	Kola Amrutha Sandhya	Reg. No.:	192524270
Section / Batch:	AIDS	Date:	24-08-2026

Code of Conduct

I Kola Amrutha Sandhya 192524270 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: Amrutha

Instructions

- This test contains 5 questions, each carrying 5 marks. Total: 25 Marks.
- No negative marking. Unanswered questions carry zero marks.
- No calculators or electronic devices permitted.
- Duration: 90 minutes.
- Graph paper or a ruler is recommended for accurate, to-scale plotting.
- Show all supporting calculations (e.g., variance explained, correlation strength, node degree) where relevant.
- Every chart must include a title, correctly labeled axes, and a legend where relevant.

Annexure A – RELATIONSHIP VISUALIZATION LAB

Rubrics for Calculation **ASSESSMENT RUBRIC** **AT1 — Relationship Visualization Lab**

CO4 · BT4: Analyze ·

This rubric is used to assess student responses to the CO4-AT1 Relationship Visualization Lab problems, in which students construct and analyze scatterplots, correlograms, PCA outputs, bubble charts, or network graphs from given data.

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Concepts	25%	Demonstrates thorough understanding of the relevant relationship-visualization technique (scatterplot, correlogram, PCA, bubble chart, or network graph) and correctly applies it to the given data.	Good understanding of the relevant technique, with only minor gaps in application.	Basic understanding, but with some misconceptions about the correct technique or how to apply it.	Poor understanding of core concepts; applies a technique fundamentally unsuited to the data.
Explanation	25%	Provides detailed, well-reasoned analysis of the constructed visualization, explicitly tied to the specific values in the dataset.	Provides clear analysis with adequate reasoning tied to the data.	Analysis is present but generic or only loosely connected to the specific dataset given.	Little or no analytical reasoning provided; answers restate the data without interpretation.
Application	20%	Effectively applies construction and analysis techniques to produce a complete, accurate	Applies techniques to produce a correct visualization, with only minor errors.	Limited application; the visualization partially represents the data but has notable errors.	Fails to apply appropriate techniques; the visualization does not correctly represent the data.

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
		visualization correctly matched to the data.			
Organization	15%	The response is clearly structured, with construction and analysis presented in a logical sequence with coherent overall flow.	The response is organized, with only minor issues in flow or clarity.	Basic structure; construction and analysis are present but the response lacks clarity or sequence.	Poorly organized; the response is incoherent or key parts are left unaddressed.
Accuracy	10%	All parts of the answer — construction, calculations, and analysis — are completely accurate and well-suited to the given data.	Mostly accurate, with only minor omissions or a small error.	Partially correct; some elements are missing, mismatched to the data, or incorrect.	Largely incorrect or inappropriate, with major gaps across multiple parts.

Scoring Guide

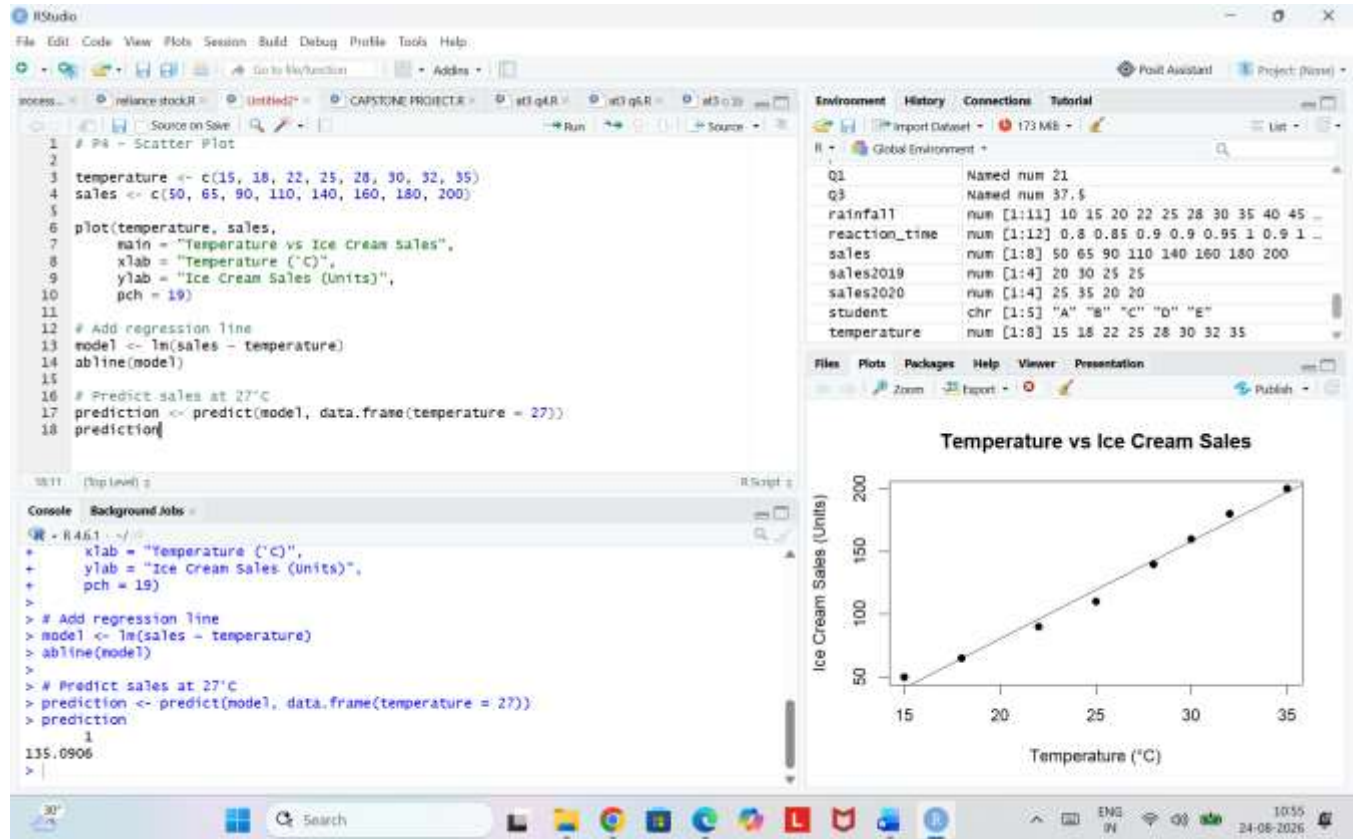
For each criterion, award a performance level (1–4) based on the descriptors above. Multiply the level achieved (as a percentage of 4, i.e., Level 4 = 100%, Level 3 = 75%, Level 2 = 50%, Level 1 = 25%) by the criterion's weightage, then sum across all five criteria to obtain the total score out of 100%.

Faculty In-charge	Course Coordinator	Head of Department
Signature: _____	Signature: _____	Signature: _____
Date: _____	Date: _____	Date: _____

Problem 4

Data: Temperature (°C) and ice cream sales (units) for 8 days: (15,50), (18,65), (22,90), (25,110), (28,140), (30,160), (32,180), (35,200).

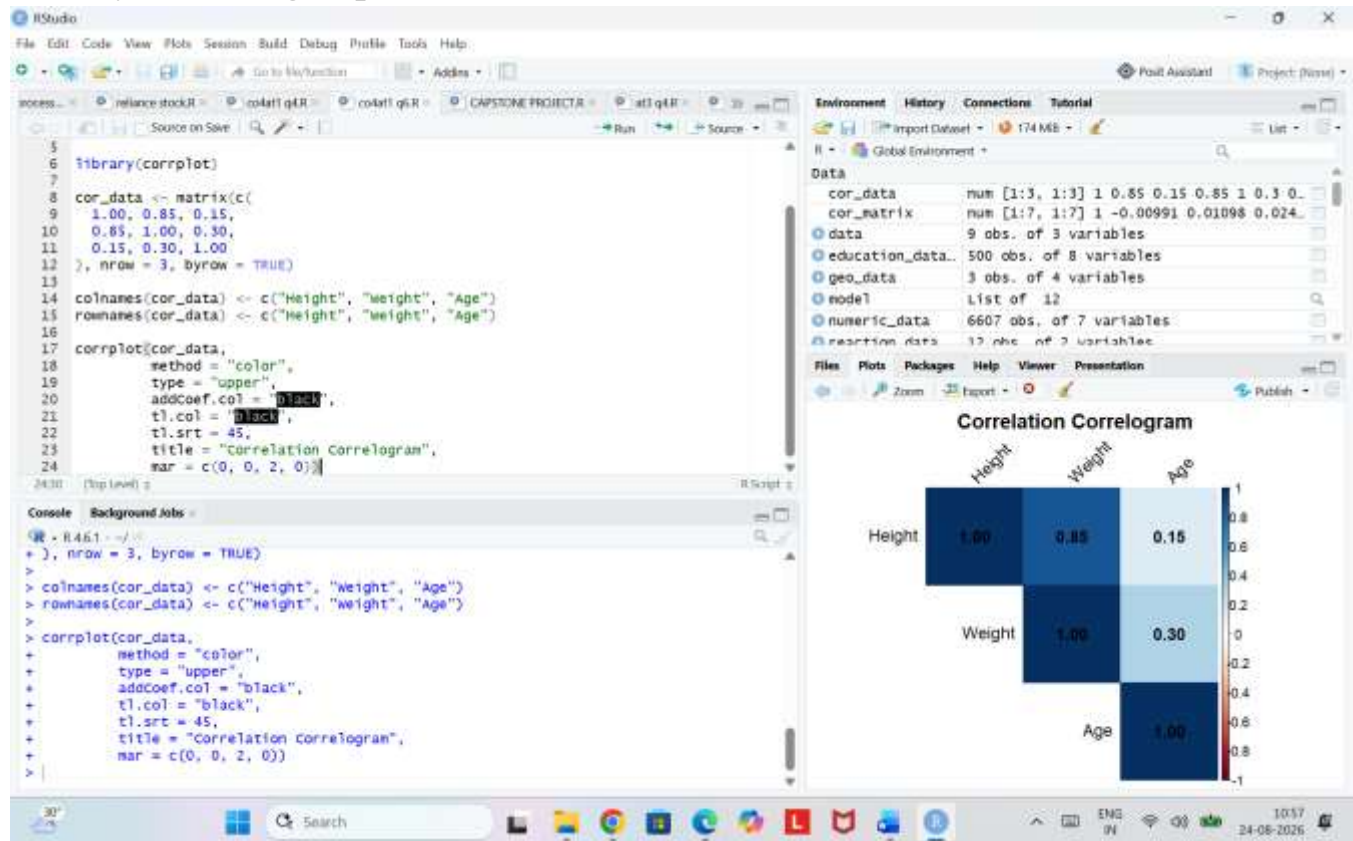
Task: Construct a scatterplot and use the pattern to predict sales at 27°C.



Problem 6

Data: Correlation values for 3 variables: Height–Weight = 0.85, Height–Age = 0.15, Weight–Age = 0.30.

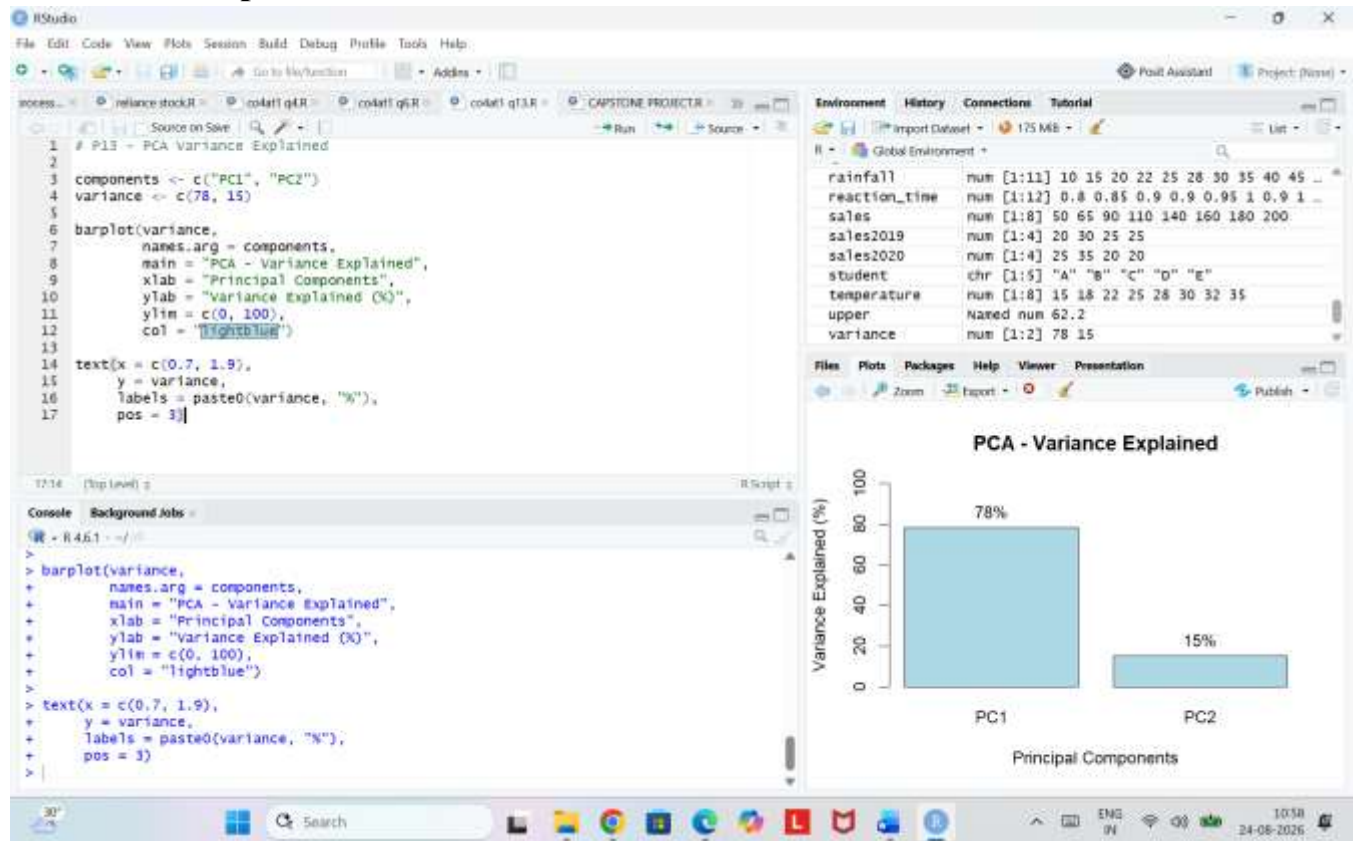
Task: Construct a correlogram (describe the shading/color for each cell) and identify the strongest pair.



Problem 13

Data: A PCA on 4 features (Revenue, Profit, Employees, Marketing Spend) shows PC1 explains 78% of variance and PC2 explains 15%.

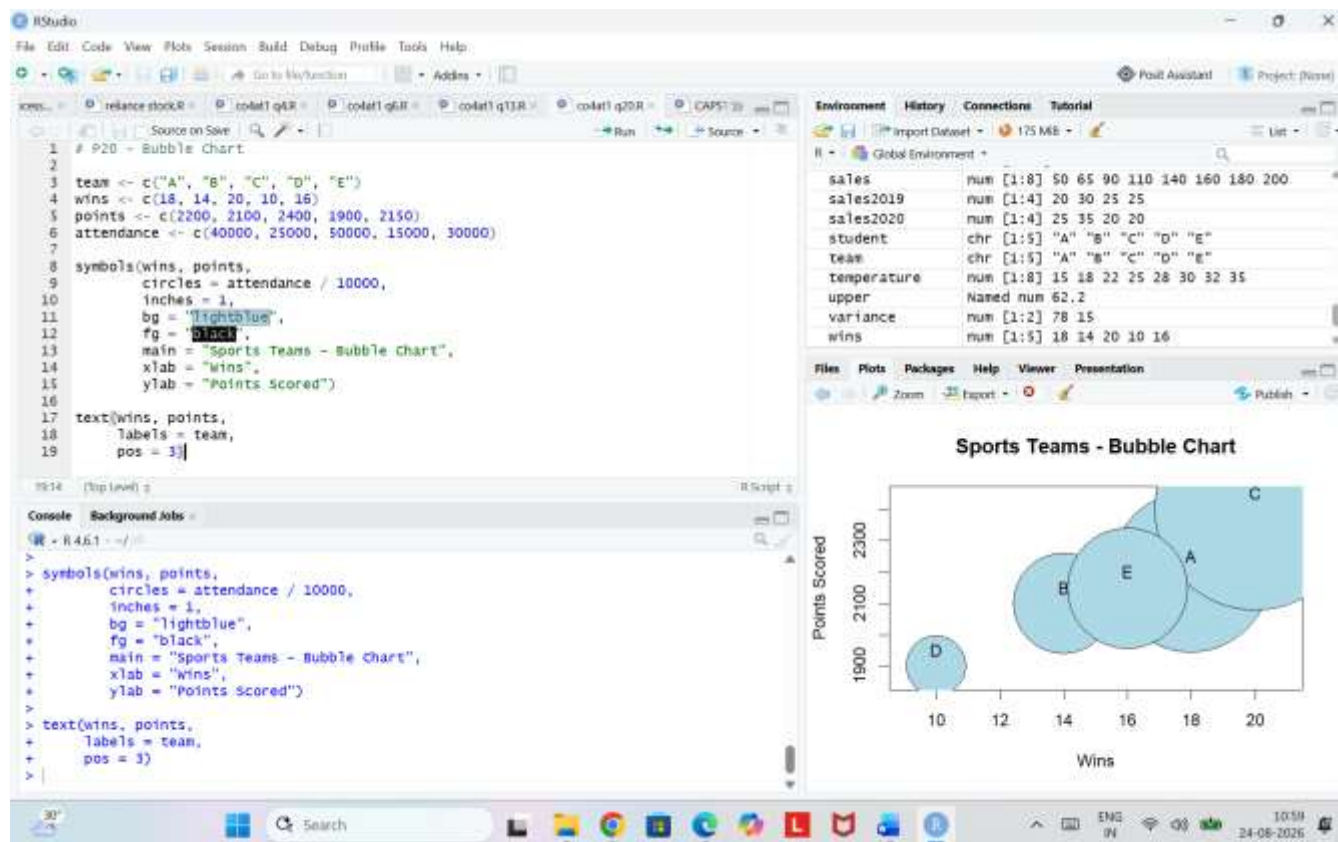
Task: Calculate the combined variance explained by PC1 + PC2, and state whether 2 components are sufficient to represent this data well.



Problem 20

Data: 5 sports teams' Wins (X), Points Scored (Y), Fan Attendance (bubble size):
A(18,2200,40000), B(14,2100,25000), C(20,2400,50000), D(10,1900,15000),
E(16,2150,30000).

Task: Construct a bubble chart and identify the team with the best overall performance based on all three variables.



Problem 22

Data: A social network of 5 friends with connections: P1–P2, P1–P3, P2–P4, P3–P4, P4–P5.

Task: Draw the network graph and calculate the degree (number of connections) of each node.

